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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Thermal Properties of Matter

JEE Main Physics · Previous Year Questions with Answer Key

120	97	2002 – 2026	16	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Numerical	17 questions
MCQ (Single Correct Answer)	87 questions
Out of syllabus	16 questions
Answer key	at the end

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Numerical 17 questions

Q1. JEE Main 2026 (Online) 5th April Evening Shift

The heat extracted out of x gram of water initially at $50^{\circ}\mathrm{C}$ to cool it down to $0^{\circ}\mathrm{C}$ is sufficient to evaporate $(1000-x)$ gram of water also initially at $50^{\circ}\mathrm{C}$. The value of x (closest integer) is ____.

(Take latent heat of water $2256\ \mathrm{kJ/kg}$, specific heat capacity of water $4200\ \mathrm{J/kg.K}$)

Open solution

Q2. JEE Main 2025 (Online) 7th April Evening Shift

Two cylindrical rods A and B made of different materials, are joined in a straight line. The ratios of lengths, radii and thermal conductivities of these rods are: $\frac{\mathrm{L_A}}{\mathrm{L_B}}=\frac{1}{2}$, $\frac{\mathrm{r_A}}{\mathrm{r_B}}=2$ and $\frac{\mathrm{K_A}}{\mathrm{K_B}}=\frac{1}{2}$. The free ends of rods A and B are maintained at 400 K , 200 K , respectively. The temperature of rods interface is _____ K, when equilibrium is established.

Open solution

Q3. JEE Main 2025 (Online) 7th April Morning Shift

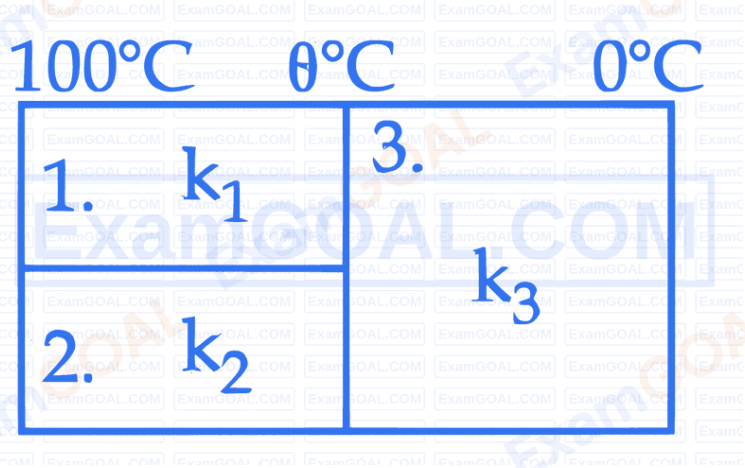
A wire of length 10 cm and diameter 0.5 mm is used in a bulb. The temperature of the wire is $1727^{\circ}\mathrm{C}$ and power radiated by the wire is 94.2 W . Its emissivity is $\frac{x}{8}$ where $x=$ _____.

(Given $\sigma=6.0\times10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}, \pi=3.14$ and assume that the emissivity of wire material is same at all wavelength.)

Open solution

Q4. JEE Main 2025 (Online) 22nd January Morning Shift

Three conductors of same length having thermal conductivity k_1, k_2 and k_3 are connected as shown in figure.



Area of cross sections of 1st and 2nd conductor are same and for 3rd conductor it is double of the 1st conductor. The temperatures are given in the figure. In steady state condition, the value of θ is _____ $^{\circ}\mathrm{C}$. (Given : $\mathrm{k_1}=60\mathrm{Js^{-1}\,m^{-1}\,K^{-1}}, \mathrm{k_2}=120\mathrm{Js^{-1}\,m^{-1}\,K^{-1}}, \mathrm{k_3}=135\mathrm{Js^{-1}\,m^{-1}\,K^{-1}}$)

Open solution

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Q5. JEE Main 2023 (Online) 13th April Evening Shift

Two plates A and B have thermal conductivities $84\ \mathrm{Wm^{-1}\,K^{-1}}$ and $126\ \mathrm{Wm^{-1}\,K^{-1}}$ respectively. They have same surface area and same thickness. They are placed in contact along their surfaces. If the temperatures of the outer surfaces of A and B are kept at $100^{\circ}\mathrm{C}$ and $0^{\circ}\mathrm{C}$ respectively, then the temperature of the surface of contact in steady state is _____ $^{\circ}\mathrm{C}$.

Open solution

Q6. JEE Main 2023 (Online) 31st January Evening Shift

A water heater of power $2000\ \mathrm{W}$ is used to heat water. The specific heat capacity of water is $4200\ \mathrm{J\ kg^{-1}\,K^{-1}}$. The efficiency of heater is 70%. Time required to heat $2\ \mathrm{kg}$ of water from $10^{\circ}\mathrm{C}$ to $60^{\circ}\mathrm{C}$ is _____ s.

(Assume that the specific heat capacity of water remains constant over the temperature range of the water).

Open solution

Q7. JEE Main 2023 (Online) 30th January Evening Shift

A faulty thermometer reads $5^{\circ}\mathrm{C}$ in melting ice and $95^{\circ}\mathrm{C}$ in steam. The correct temperature on absolute scale will be _____ K when the faulty thermometer reads $41^{\circ}\mathrm{C}$.

Open solution

Q8. JEE Main 2023 (Online) 24th January Morning Shift

A hole is drilled in a metal sheet. At $27^{\circ}\mathrm{C}$, the diameter of hole is 5 cm. When the sheet is heated to $177^{\circ}\mathrm{C}$, the change in the diameter of hole is $\mathrm{d} \times 10^{-3}$ cm. The value of d will be _____ if coefficient of linear expansion of the metal is $1.6 \times 10^{-5}/^{\circ}\mathrm{C}$.

Open solution

Q9. JEE Main 2022 (Online) 25th July Evening Shift

A block of ice of mass 120 g at temperature 0°C is put in 300 g of water at 25°C. The x g of ice melts as the temperature of the water reaches 0°C. The value of x is _____.

[Use specific heat capacity of water = 4200 $\mathrm{Jkg^{-1}K^{-1}}$, Latent heat of ice = 3.5×10^5 $\mathrm{Jkg^{-1}}$]

Open solution

Q10. JEE Main 2022 (Online) 25th July Morning Shift

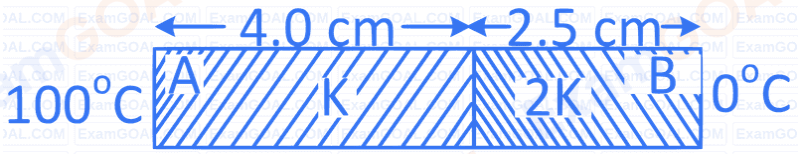
A unit scale is to be prepared whose length does not change with temperature and remains $20\ \mathrm{cm}$, using a bimetallic strip made of brass and iron each of different length. The length of both components would change in such a way that difference between their lengths remains constant. If length of brass is $40\ \mathrm{cm}$, then length of iron will be _____ cm. $\left(\alpha_{\text{iron}} = 1.2 \times 10^{-5}\ \mathrm{K^{-1}}\right.$ and $\left.\alpha_{\text{brass}} = 1.8 \times 10^{-5}\ \mathrm{K^{-1}}\right)$.

Open solution

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Q11. JEE Main 2022 (Online) 29th June Morning Shift

As per the given figure, two plates A and B of thermal conductivity K and 2 K are joined together to form a compound plate. The thickness of plates are 4.0 cm and 2.5 cm respectively and the area of cross-section is 120 cm^2 for each plate. The equivalent thermal conductivity of the compound plate is $\left(1+\frac{5}{\alpha}\right)$ K, then the value of α will be _____.



Open solution

Q12. JEE Main 2022 (Online) 26th June Evening Shift

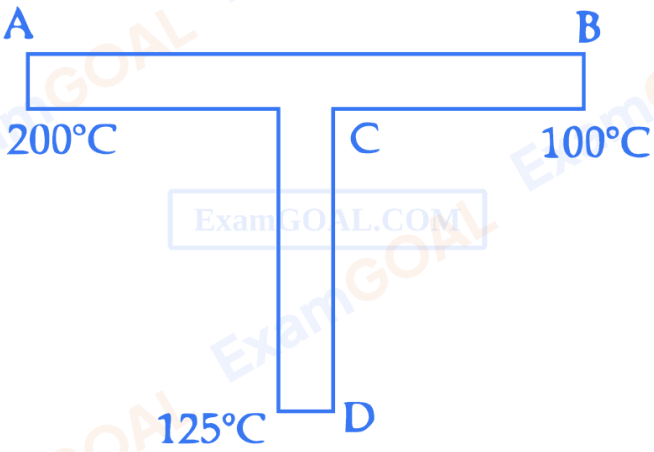
A geyser heats water flowing at a rate of 2.0 kg per minute from 30°C to 70°C. If geyser operates on a gas burner, the rate of combustion of fuel will be _____ g min^{-1}.

[Heat of combustion = 8×10^3 Jg^{-1}, Specific heat of water = 4.2 Jg^{-1} $°C^{-1}$]

Open solution

Q13. JEE Main 2021 (Online) 27th August Morning Shift

A rod CD of thermal resistance 10.0 KW^{-1} is joined at the middle of an identical rod AB as shown in figure. The end A, B and D are maintained at 200°C, 100°C and 125°C respectively. The heat current in CD is P watt. The value of P is ..... .



Open solution

Q14. JEE Main 2020 (Online) 3rd September Morning Slot

A bakelite beaker has volume capacity of 500 cc at 30°C. When it is partially filled with Vm volume (at 30°C) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(beaker)} = 6 \times 10^{-6}$ $°C^{-1}$ and $\gamma_{(mercury)} = 1.5 \times 10^{-4}$ $°C^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to ____.

Open solution

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Q15. JEE Main 2020 (Online) 8th January Evening Slot

Three containers C_1, C_2 and C_3 have water at different temperatures. The table below shows the final temperature T when different amounts of water (given in litres) are taken from each containers and mixed (assume no loss of heat during the process)

Open solution

C_1	C_2	C_3	T
$1l$	$2l$	–	60°C
–	$1l$	$2l$	30°C
$2l$	–	$1l$	60°C
$1l$	$1l$	$1l$	θ

The value of θ (in °C to the nearest integer) is .....

Q16. JEE Main 2020 (Online) 7th January Evening Slot

M grams of steam at 100°C is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produced liquid water at 40°C [heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g] the value of M is ____

Open solution

Q17. JEE Main 2020 (Online) 7th January Morning Slot

A non-isotropic solid metal cube has coefficients of linear expansion as :
5×10^{-5}/°C along the x-axis and 5×10^{-6}/°C along the y and the z-axis. If the coefficient of volume expansion of the solid is C $\times 10^{-6}$/°C then the value of C is

Open solution

MCQ (Single Correct Answer) 87 questions

Q18. JEE Main 2026 (Online) 4th April Evening Shift

The temperature of a metal strip having coefficient of linear expansion α is increased from T_1 to T_2 resulting in increase of its length by ΔL_1. The temperature is further increased from T_2 to T_3 such that the increase in its length is ΔL_2.

Given $T_3 + T_1 = 2T_2$ and $T_2 - T_1 = \Delta T$, the value of ΔL_2 is ____ .

Open solution

(A) $\Delta L_1 \left[1 + 2\alpha^2(\Delta T)^2\right]$

(B) $\Delta L_1 \left[1 + \alpha^2(\Delta T)^2\right]$

(C) $\Delta L_1[1 + 2\alpha\Delta T]$

(D) $\Delta L_1[1 + \alpha\Delta T]$

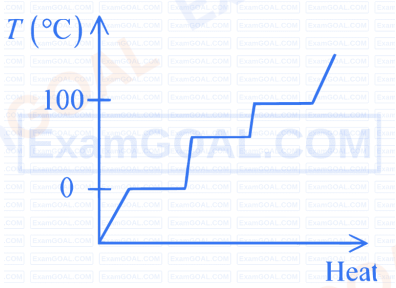
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Q19. JEE Main 2026 (Online) 28th January Morning Shift

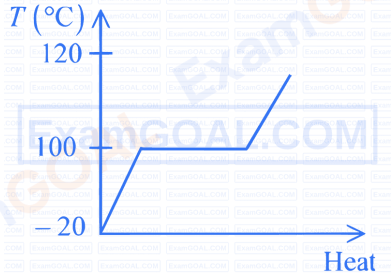
Which of the following best represents the temperature versus heat supplied graph for water, in the range of $-20^{\circ}\mathrm{C}$ to $120^{\circ}\mathrm{C}$?

Open solution

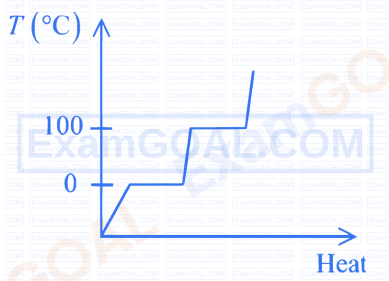
(A)



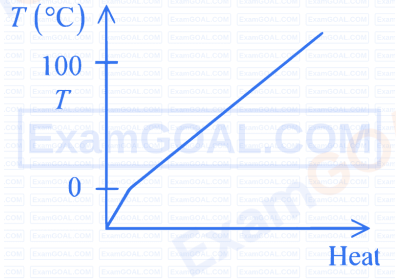
(B)



(C)



(D)



Q20. JEE Main 2026 (Online) 28th January Morning Shift

10 kg of ice at $-10^{\circ}\mathrm{C}$ is added to 100 kg of water to lower its temperature from 25 $^{\circ}\mathrm{C}$. Consider no heat exchange to surroundings. The decrement to the temperature of water is ____ $^{\circ}\mathrm{C}$.

(specific heat of ice $= 2100\ \mathrm{J/Kg.^{\circ}C}$, specific heat of water $= 4200\ \mathrm{J/Kg.^{\circ}C}$, latent heat of fusion of ice $= 3.36 \times 10^5\ \mathrm{J/Kg}$)

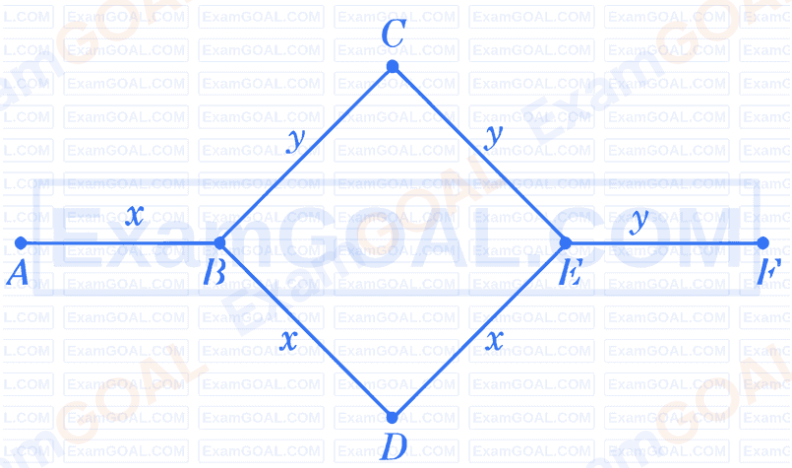
Open solution

(A) 15 (B) 10 (C) 6.67 (D) 11.6

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Q21. JEE Main 2026 (Online) 22nd January Morning Shift

Rods x and y of equal dimensions but of different materials are joined as shown in figure. Temperatures of end points A and F are maintained at 100°C and 40°C respectively. Given the thermal conductivity of $\text{rod}\ x$ is three times of that of $\text{rod}\ y$, the temperature at junction points B and E are (close to):



Open solution

(A) 60°C and 45°C respectively

(B) 80°C and 70°C respectively

(C) 89°C and 73°C respectively

(D) 80°C and 60°C respectively

Q22. JEE Main 2026 (Online) 21st January Morning Shift

An aluminium and steel rods having same lengths and cross-sections are joined to make total length of 120 cm at 30°C. The coefficient of linear expansion of aluminium and steel are $24 \times 10^{-6}/^\circ\text{C}$ and $1.2 \times 10^{-5}/^\circ\text{C}$, respectively. The length of this composite rod when its temperature is raised to 100°C, is ____ cm.

Open solution

(A) 120.20 (B) 120.06 (C) 120.15 (D) 120.03

Q23. JEE Main 2026 (Online) 21st January Morning Shift

A gas based geyser heats water flowing at the rate of 5.0 litres per minute from 27°C to 87°C. The rate of consumption of the gas is ____ g/s.

(Take heat of combustion of gas $= 5.0 \times 10^4\ \text{J/g}$, specific heat capacity of water $= 4200\ \text{J/kg} \cdot {}^\circ\text{C}$)

Open solution

(A) 4.2 (B) 2.1 (C) 0.21 (D) 0.42

Q24. JEE Main 2025 (Online) 8th April Evening Shift

Water falls from a height of 200 m into a pool. Calculate the rise in temperature of the water assuming no heat dissipation from the water in the pool.

(Take g = 10 m/s^2, specific heat of water = 4200 J/(kg K))

Open solution

(A) 0.36 K (B) 0.23 K (C) 0.48 K (D) 0.14 K

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Q25. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Consider a rectangular sheet of solid material of length $l = 9\ \text{cm}$ and width $\text{d} = 4\ \text{cm}$. The coefficient of linear expansion is $\alpha = 3.1 \times 10^{-5}\ \text{K}^{-1}$ at room temperature and one atmospheric pressure. The mass of sheet $m = 0.1\ \text{kg}$ and the specific heat capacity $C_\text{v} = 900\ \text{J kg}^{-1}\ \text{K}^{-1}$. If the amount of heat supplied to the material is $8.1 \times 10^2\ \text{J}$ then change in area of the rectangular sheet is :

(A) $2.0 \times 10^{-6}\ \text{m}^2$ (B) $6.0 \times 10^{-7}\ \text{m}^2$ (C) $3.0 \times 10^{-7}\ \text{m}^2$ (D) $4.0 \times 10^{-7}\ \text{m}^2$

Q26. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Match List - I with List - II.

List - I		$List - II$
(A)	Heat capacity of body	(I) J kg^{-1}
(B)	Specific heat capacity of body	(II) J K^{-1}
(C)	Latent heat	(III) $\text{J kg}^{-1}\ \text{K}^{-1}$
(D)	Thermal conductivity	(IV) $\text{J m}^{-1}\ \text{K}^{-1}\ \text{s}^{-1}$

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(B) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
(D) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Q27. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

The difference of temperature in a material can convert heat energy into electrical energy. To harvest the heat energy, the material should have

(A) low thermal conductivity and high electrical conductivity
(B) low thermal conductivity and low electrical conductivity
(C) high thermal conductivity and high electrical conductivity
(D) high thermal conductivity and low electrical conductivity

Q28. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

A cup of coffee cools from 90°C to 80°C in t minutes when the room temperature is 20°C. The time taken by the similar cup of coffee to cool from 80°C to 60°C at the same room temperature is:

(A) $\frac{13}{5}t$ (B) $\frac{10}{13}t$ (C) $\frac{5}{13}t$ (D) $\frac{13}{10}t$

Q29. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

The temperature of a body in air falls from 40°C to 24°C in 4 minutes. The temperature of the air is 16°C. The temperature of the body in the next 4 minutes will be :

(A) $\frac{28}{3}^\circ\text{C}$ (B) $\frac{56}{3}^\circ\text{C}$
(C) $\frac{42}{3}^\circ\text{C}$ (D) $\frac{14}{3}^\circ\text{C}$

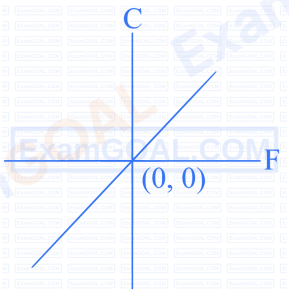
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Q30. JEE Main 2025 (Online) 24th January Evening Shift

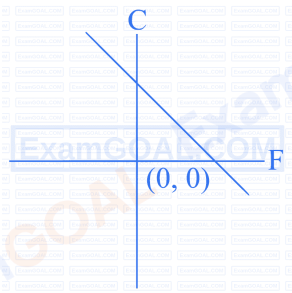
Open solution

Which of the following figure represents the relation between Celsius and Fahrenheit temperatures?

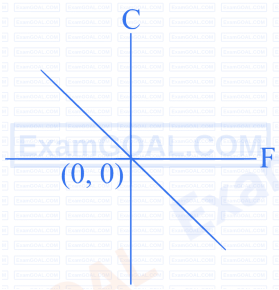
(A)



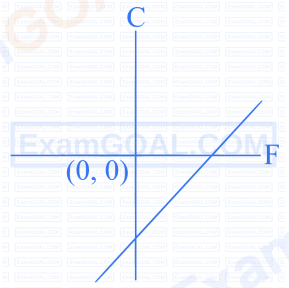
(B)



(C)



(D)



Q31. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

A gun fires a lead bullet of temperature 300 K into a wooden block. The bullet having melting temperature of 600 K penetrates into the block and melts down. If the total heat required for the process is 625 J , then the mass of the bullet is _____ grams. (Latent heat of fusion of lead $= 2.5 \times 10^4 \mathrm{JKg}^{-1}$ and specific heat capacity of lead $= 125 \mathrm{JKg}^{-1}\,\mathrm{K}^{-1}$)

(A) 20 (B) 15 (C) 5 (D) 10

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Q32. JEE Main 2025 (Online) 22nd January Morning Shift

Two spherical bodies of same materials having radii 0.2 m and 0.8 m are placed in same atmosphere. The temperature of the smaller body is 800 K and temperature of the bigger body is 400 K . If the energy radiated from the smaller body is E, the energy radiated from the bigger body is (assume, effect of the surrounding temperature to be negligible),

Open solution

(A) 64 E (B) 16 E (C) E (D) 256 E

Q33. JEE Main 2025 (Online) 22nd January Morning Shift

An amount of ice of mass $10^{-3}\ \mathrm{kg}$ and temperature $-10^{\circ}\mathrm{C}$ is transformed to vapour of temperature $110^{\circ}\mathrm{C}$ by applying heat. The total amount of work required for this conversion is, (Take, specific heat of ice $= 2100\mathrm{Jkg}^{-1}\ \mathrm{K}^{-1}$, specific heat of water $= 4180\mathrm{Jkg}^{-1}\ \mathrm{K}^{-1}$, specific heat of steam $= 1920\mathrm{Jkg}^{-1}\ \mathrm{K}^{-1}$, Latent heat of ice $= 3.35 \times 10^{5}\mathrm{Jkg}^{-1}$ and Latent heat of steam $= 2.25 \times 10^{6}\ \mathrm{Jkg}^{-1}$)

Open solution

(A) 3022 J (B) 3043 J (C) 3003 J (D) 3024 J

Q34. JEE Main 2024 (Online) 4th April Morning Shift

The resistances of the platinum wire of a platinum resistance thermometer at the ice point and steam point are 8Ω and 10Ω respectively. After inserting in a hot bath of temperature $400^{\circ}\mathrm{C}$, the resistance of platinum wire is :

Open solution

(A) 10 Ω (B) 16 Ω (C) 8 Ω (D) 2 Ω

Q35. JEE Main 2024 (Online) 4th April Morning Shift

On Celsius scale the temperature of body increases by $40^{\circ}\mathrm{C}$. The increase in temperature on Fahrenheit scale is :

Open solution

(A) $75^{\circ}\mathrm{F}$ (B) $70^{\circ}\mathrm{F}$ (C) $72^{\circ}\mathrm{F}$ (D) $68^{\circ}\mathrm{F}$

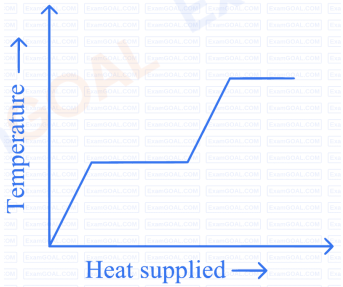
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Q36. JEE Main 2024 (Online) 30th January Evening Shift

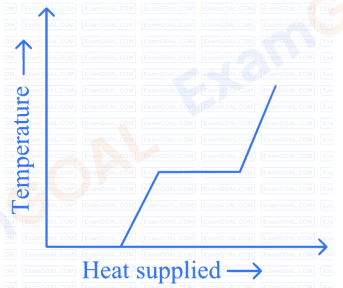
Open solution

A block of ice at $-10^{\circ}\mathrm{C}$ is slowly heated and converted to steam at $100^{\circ}\mathrm{C}$. Which of the following curves represent the phenomenon qualitatively:

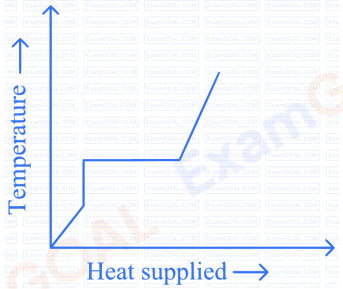
(A)



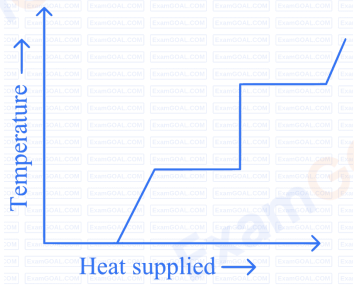
(B)



(C)



(D)



Q37. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

On a temperature scale 'X', the boiling point of water is $65^{\circ}\mathrm{X}$ and the freezing point is $-15^{\circ}\mathrm{X}$. Assume that the X scale is linear. The equivalent temperature corresponding to $-95^{\circ}\mathrm{X}$ on the Fahrenheit scale would be:

(A) $-148^{\circ}\mathrm{F}$ (B) $-48^{\circ}\mathrm{F}$ (C) $-63^{\circ}\mathrm{F}$ (D) $-112^{\circ}\mathrm{F}$

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Q38. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Heat energy of 184 kJ is given to ice of mass 600 g at $-12^{\circ}C$. Specific heat of ice is $2222.3\ J\ kg^{-1^{\circ}}\ C^{-1}$ and latent heat of ice is 336 $kJ\ kg^{-1}$

A. Final temperature of system will be $0^{\circ}C$.

B. Final temperature of the system will be greater than $0^{\circ}C$.

C. The final system will have a mixture of ice and water in the ratio of 5 : 1.

D. The final system will have a mixture of ice and water in the ratio of 1 : 5.

E. The final system will have water only.

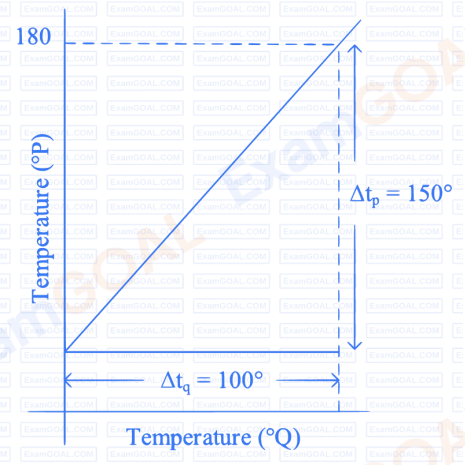
Choose the correct answer from the options given below :

(A) A and E only (B) B and D only (C) A and C only (D) A and D only

Q39. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

The graph between two temperature scales P and Q is shown in the figure. Between upper fixed point and lower fixed point there are 150 equal divisions of scale P and 100 divisions on scale Q. The relationship for conversion between the two scales is given by :-



(A) $\frac{t_P}{100} = \frac{t_Q - 180}{150}$ (B) $\frac{t_P}{180} = \frac{t_Q - 40}{100}$

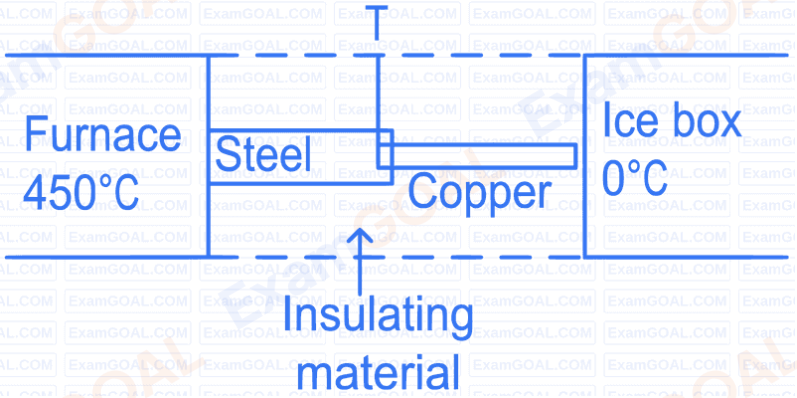
(C) $\frac{t_Q}{150} = \frac{t_P - 180}{100}$ (D) $\frac{t_Q}{100} = \frac{t_P - 30}{150}$

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Q40. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

If K_1 and K_2 are the thermal conductivities, L_1 and L_2 are the lengths and A_1 and A_2 are the cross sectional areas of steel and copper rods respectively such that $\frac{K_2}{K_1} = 9$, $\frac{A_1}{A_2} = 2$, $\frac{L_1}{L_2} = 2$. Then, for the arrangement as shown in the figure, the value of temperature T of the steel - copper junction in the steady state will be:



(A) $18^\circ C$ (B) $14^\circ C$ (C) $45^\circ C$ (D) $150^\circ C$

Q41. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

Read the following statements :

A. When small temperature difference between a liquid and its surrounding is doubled, the rate of loss of heat of the liquid becomes twice.

B. Two bodies P and Q having equal surface areas are maintained at temperature $10^\circ C$ and $20^\circ C$. The thermal radiation emitted in a given time by P and Q are in the ratio $1 : 1.15$.

C. A Carnot Engine working between $100\ K$ and $400\ K$ has an efficiency of 75%.

D. When small temperature difference between a liquid and its surrounding is quadrupled, the rate of loss of heat of the liquid becomes twice.

Choose the correct answer from the options given below :

(A) A, B, C only (B) A, B only (C) A, C only (D) B, C, D only

Q42. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

An ice cube of dimensions $60\ cm \times 50\ cm \times 20\ cm$ is placed in an insulation box of wall thickness $1\ cm$. The box keeping the ice cube at $0^\circ C$ of temperature is brought to a room of temperature $40^\circ C$. The rate of melting of ice is approximately :

(Latent heat of fusion of ice is $3.4 \times 10^5\ J\ kg^{-1}$ and thermal conducting of insulation wall is $0.05\ Wm^{-1}{}^\circ C^{-1}$)

(A) $61 \times 10^{-3}\ kg\ s^{-1}$ (B) $61 \times 10^{-5}\ kg\ s^{-1}$

(C) $208\ kg\ s^{-1}$ (D) $30 \times 10^{-5}\ kg\ s^{-1}$

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Q43. JEE Main 2022 (Online) 30th June Morning Shift

The pressure of the gas in a constant volume gas thermometer is 100 cm of mercury when placed in melting ice at 1 atm. When the bulb is placed in a liquid, the pressure becomes 180 cm of mercury. Temperature of the liquid is :

(Given 0°C = 273 K)

(A) 300 K (B) 400 K (C) 600 K (D) 491 K

Open solution

Q44. JEE Main 2022 (Online) 29th June Evening Shift

At what temperature a gold ring of diameter 6.230 cm be heated so that it can be fitted on a wooden bangle of diameter 6.241 cm ? Both the diameters have been measured at room temperature (27°C).

(Given : coefficient of linear thermal expansion of gold $\alpha_L = 1.4 \times 10^{-5}\ K^{-1}$)

(A) 125.7°C (B) 91.7°C (C) 425.7°C (D) 152.7°C

Open solution

Q45. JEE Main 2022 (Online) 27th June Evening Shift

A lead bullet penetrates into a solid object and melts. Assuming that 40% of its kinetic energy is used to heat it, the initial speed of bullet is :

(Given : initial temperature of the bullet = 127°C, Melting point of the bullet = 327°C, Latent heat of fusion of lead = $2.5 \times 10^4\ J\ kg^{-1}$, Specific heat capacity of lead = 125 J/kg K)

(A) $125\ ms^{-1}$ (B) $500\ ms^{-1}$ (C) $250\ ms^{-1}$ (D) $600\ ms^{-1}$

Open solution

Q46. JEE Main 2022 (Online) 25th June Evening Shift

A solid metallic cube having total surface area 24 m^2 is uniformly heated. If its temperature is increased by 10°C, calculate the increase in volume of the cube. (Given $\alpha = 5.0 \times 10^{-4}\ °C^{-1}$).

(A) $2.4 \times 10^6\ cm^3$ (B) $1.2 \times 10^5\ cm^3$ (C) $6.0 \times 10^4\ cm^3$ (D) $4.8 \times 10^5\ cm^3$

Open solution

Q47. JEE Main 2022 (Online) 25th June Evening Shift

A copper block of mass 5.0 kg is heated to a temperature of 500°C and is placed on a large ice block. What is the maximum amount of ice that can melt? [Specific heat of copper : 0.39 $J\ g^{-1}\ °C^{-1}$ and latent heat of fusion of water : 335 $J\ g^{-1}$]

(A) 1.5 kg (B) 5.8 kg (C) 2.9 kg (D) 3.8 kg

Open solution

Q48. JEE Main 2022 (Online) 24th June Evening Shift

A 100 g of iron nail is hit by a 1.5 kg hammer striking at a velocity of 60 ms^{-1}. What will be the rise in the temperature of the nail if one fourth of energy of the hammer goes into heating the nail?

[Specific heat capacity of iron = 0.42 $Jg^{-1}\ °C^{-1}$]

(A) 675°C (B) 1600°C (C) 16.07°C (D) 6.75°C

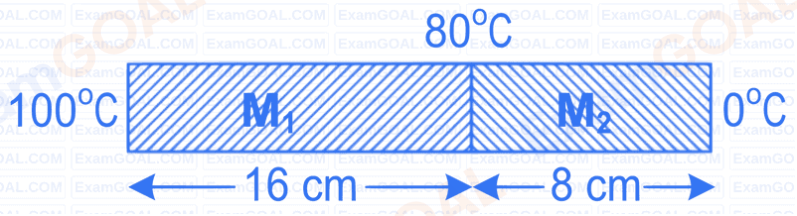
Open solution

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Q49. JEE Main 2022 (Online) 24th June Morning Shift

Two metallic blocks M_1 and M_2 of same area of cross-section are connected to each other (as shown in figure). If the thermal conductivity of M_2 is K then the thermal conductivity of M_1 will be :

[Assume steady state heat conduction]



(A) 10 K (B) 8 K (C) 12.5 K (D) 2 K

Open solution

Q50. JEE Main 2021 (Online) 31st August Evening Shift

Two thin metallic spherical shells of radii r_1 and r_2 ($r_1 < r_2$) are placed with their centres coinciding. A material of thermal conductivity K is filled in the space between the shells. The inner shell is maintained at temperature θ_1 and the outer shell at temperature $\theta_2(\theta_1 < \theta_2)$. The rate at which heat flows radially through the material is :-

(A) $\frac{4\pi K r_1 r_2(\theta_2-\theta_1)}{r_2-r_1}$ (B) $\frac{\pi r_1 r_2(\theta_2-\theta_1)}{r_2-r_1}$

(C) $\frac{K(\theta_2-\theta_1)}{r_2-r_1}$ (D) $\frac{K(\theta_2-\theta_1)(r_2-r_1)}{4\pi r_1 r_2}$

Open solution

Q51. JEE Main 2021 (Online) 27th August Evening Shift

The height of Victoria falls is 63 m. What is the difference in temperature of water at the top and at the bottom of fall?

[Given 1 cal = 4.2 J and specific heat of water = 1 cal $g^{-1}\,{}^\circ C^{-1}$]

(A) 0.147° C (B) 14.76° C (C) 1.476° C (D) 0.014° C

Open solution

Q52. JEE Main 2021 (Online) 26th August Evening Shift

The temperature of equal masses of three different liquids x, y and z are 10°C, 20°C and 30°C respectively. The temperature of mixture when x is mixed with y is 16°C and that when y is mixed with z is 26°C. The temperature of mixture when x and z are mixed will be :

(A) 28.32°C (B) 25.62°C (C) 23.84°C (D) 20.28°C

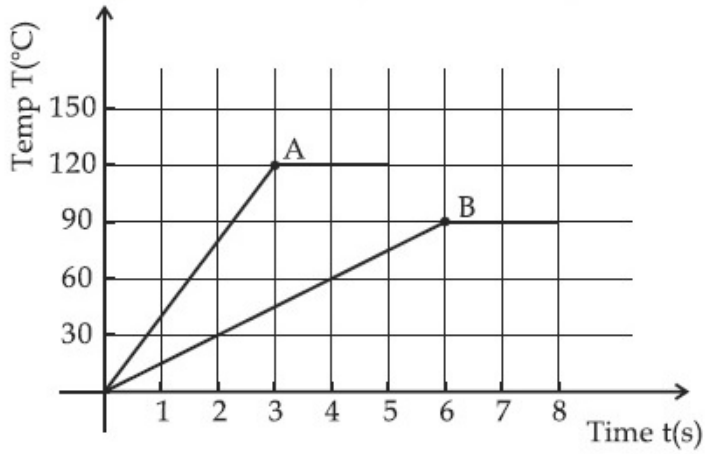
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Q53. JEE Main 2021 (Online) 25th July Morning Shift

Two different metal bodies A and B of equal mass are heated at a uniform rate under similar conditions. The variation of temperature of the bodies is graphically represented as shown in the figure. The ratio of specific heat capacities is :

Open solution



(A) $\frac{8}{3}$ (B) $\frac{3}{8}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$

Q54. JEE Main 2021 (Online) 17th March Morning Shift

Two identical metal wires of thermal conductivities K_1 and K_2 respectively are connected in series. The effective thermal conductivity of the combination is :

Open solution

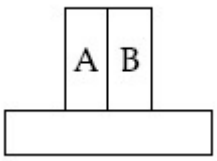
(A) $\frac{2K_1K_2}{K_1+K_2}$ (B) $\frac{K_1+K_2}{K_1K_2}$

(C) $\frac{K_1+K_2}{2K_1K_2}$ (D) $\frac{K_1K_2}{K_1+K_2}$

Q55. JEE Main 2021 (Online) 16th March Evening Shift

A bimetallic strip consists of metals A and B. It is mounted rigidly as shown. The metal A has higher coefficient of expansion compared to that of metal B. When the bimetallic strip is placed in a cold bath, it will :

Open solution



(A) Neither bend nor shrink (B) Bend towards the left

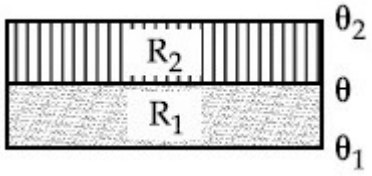
(C) Not bend but shrink (D) Bend towards the right

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Q56. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

The temperature θ at the junction of two insulating sheets, having thermal resistances R_1 and R_2 as well as top and bottom temperatures θ_1 and θ_2 (as shown in figure) is given by :



(A) $\frac{\theta_1 R_2+\theta_2 R_1}{R_1+R_2}$

(B) $\frac{\theta_1 R_1+\theta_2 R_2}{R_1+R_2}$

(C) $\frac{\theta_1 R_2-\theta_2 R_1}{R_2-R_1}$

(D) $\frac{\theta_2 R_2-\theta_1 R_1}{R_2-R_1}$

Q57. JEE Main 2021 (Online) 24th February Morning Shift

Open solution

Each side of a box made of metal sheet in cubic shape is 'a' at room temperature 'T', the coefficient of linear expansion of the metal sheet is 'α'. The metal sheet is heated uniformly, by a small temperature ΔT, so that its new temperature is T + ΔT. Calculate the increase in the volume of the metal box.

(A) $3a^3\alpha\Delta T$

(B) $4\pi a^3\alpha\Delta T$

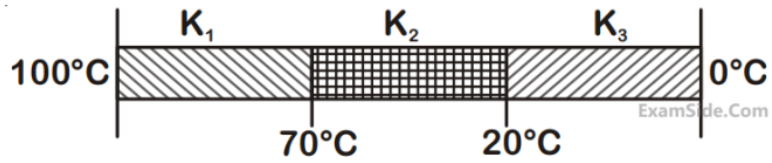
(C) $\frac{4}{3}\pi a^3\alpha\Delta T$

(D) $4a^3\alpha\Delta T$

Q58. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity K_1 , K_2 and K_3 , respectively. They are joined together at their ends to make a long rod (see figure). One end of the long rod is maintained at 100°C and the other at 0°C (see figure). If the joints of the rod are at 70°C and 20°C in steady state and there is no loss of energy from the surface of the rod, the correct relationship between K_1 , K_2 and K_3 is :



(A) $K_1 : K_3 = 2 : 3$, $K_2 : K_3 = 2 : 5$

(B) $K_1 < K_2 < K_3$

(C) $K_1 : K_2 = 5 : 2$, $K_1 : K_3 = 3 : 5$

(D) $K_1 > K_2 > K_3$

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Q59. JEE Main 2020 (Online) 5th September Evening Slot

Two different wires having lengths L_1 and L_2, and respective temperature coefficient of linear expansion α_1 and α_2, are joined end-to-end. Then the effective temperature coefficient of linear expansion is :

Open solution

(A) $2\sqrt{\alpha_1\alpha_2}$

(B) $4\frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}\frac{L_2L_1}{(L_2+L_1)^2}$

(C) $\frac{\alpha_1+\alpha_2}{2}$

(D) $\frac{\alpha_1L_1+\alpha_2L_2}{L_1+L_2}$

Q60. JEE Main 2020 (Online) 5th September Morning Slot

A bullet of mass 5 g, travelling with a speed of 210 m/s, strikes a fixed wooden target. One half of its kinetic energy is converted into heat in the bullet while the other half is converted into heat in the wood. The rise of temperature of the bullet if the specific heat of its material is 0.030 cal/(g – ^oC) (1 cal = 4.2 × 10^7 ergs) close to :

Open solution

(A) 87.5 ^oC (B) 83.3 ^oC (C) 38.4 ^oC (D) 119.2 ^oC

Q61. JEE Main 2020 (Online) 4th September Morning Slot

The specific heat of water = 4200 J $kg^{-1}K^{-1}$ and the latent heat of ice = 3.4 $\times$ 10^5 J kg^{-1}. 100 grams of ice at 0^oC is placed in 200 g of water at 25^oC. The amount of ice that will melt as the temperature of water reaches 0^oC is close to (in grams) :

Open solution

(A) 63.8 (B) 61.7 (C) 69.3 (D) 64.6

Q62. JEE Main 2020 (Online) 3rd September Evening Slot

A calorimeter of water equivalent 20 g contains 180 g of water at 25^oC. 'm' grams of steam at 100^oC is mixed in it till the temperature of the mixure is 31^oC. The value of 'm' is close to : (Latent heat of water = 540 cal g^{-1}, specific heat of water = 1 cal g^{-1} $^oC^{-1}$)

Open solution

(A) 2.6 (B) 2 (C) 4 (D) 3.2

Q63. JEE Main 2020 (Online) 2nd September Evening Slot

When the temperature of a metal wire is increased from 0^oC to 10^oC, its length increases by 0.02%. The percentage change in its mass density will be closest to :

Open solution

(A) 0.008 (B) 0.06 (C) 0.8 (D) 2.3

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Q64. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

One kg of water, at 20°C, heated in an electric kettle whose heating element has a mean (temperature averaged) resistance of 20 Ω. The rms voltage in the mains is 200 V. Ignoring heat loss from the kettle, time taken for water to evaporate fully, is close to : [Specific heat of water = 4200 J/(kg °C), Latent heat of water = 2260 kJ/kg]

(A) 10 minutes (B) 22 minutes (C) 3 minutes (D) 16 minutes

Q65. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

When M_1 gram of ice at −10°C (specific heat = 0.5 cal g^{-1} $^oC^{-1}$) is added to M_2 gram of water at 50°C, finally no ice is left and the water is at 0°C. The value of latent heat of ice, in cal g^{-1} is :

(A) $\frac{50M_2}{M_1} - 5$ (B) $\frac{50M_2}{M_1}$ (C) $\frac{5M_2}{M_1} - 5$ (D) $\frac{5M_1}{M_2} - 50$

Q66. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

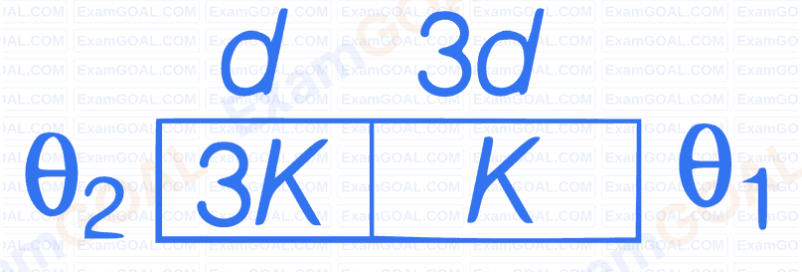
A massless spring (k = 800 N/m), attached with a mass (500 g) is completely immersed in 1 kg of water. The spring is stretched by 2 cm and released so that it starts vibrating. What would be the order of magnitude of the change in the temperature of water when the vibrations stop completely ? (Assume that the water container and spring receive negligible heat and specific heat of mass = 400 J/kg K, specific heat of water = 4184 J/kg K)

(A) 10^{-3} K (B) 10^{-1} K (C) 10^{-5}K (D) 10^{-4} K

Q67. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

Two materials having coefficients of thermal conductivity '3K' and 'K' and thickness 'd' and '3d', respectively, are joined to form a slab as shown in the figure. The temperatures of the outer surfaces are 'θ_2' and 'θ_1' respectively, ($\theta_2 > \theta_1$). The temperature at the interface is :-



(A) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$ (B) $\frac{\theta_2+\theta_1}{2}$
(C) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$ (D) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

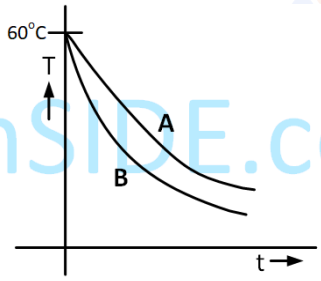
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Q68. JEE Main 2019 (Online) 8th April Morning Slot

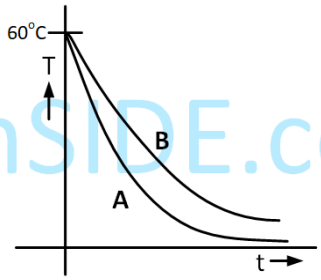
Open solution

Two identical beakers A and B contain equal volumes of two different liquids at 60°C each and left to cool down. Liquid in A has density of 8×10^2 kg/m^3 and specific heat of 2000 J kg^{-1} K^{-1} while liquid in B has density of 10^3 kg m^{-3} and specific heat of 4000 J kg^{-1} K^{-1}. Which of the following best describes their temperature versus time graph schematically? (assume the emissivity of both the beakers to be the same)

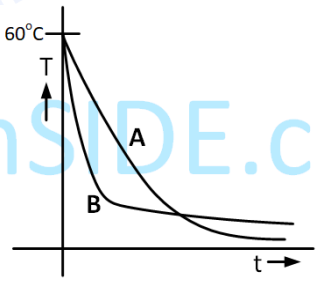
(A)



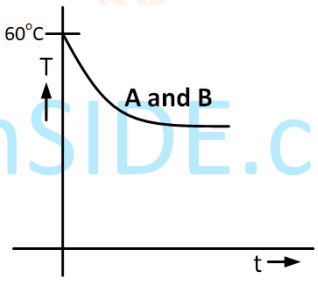
(B)



(C)



(D)



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Q69. JEE Main 2019 (Online) 8th April Morning Slot

Open solution

A thermally insulated vessel contains 150g of water at 0°C. Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be closest to : (Latent heat of vaporization of water = 2.10×10^6 J kg^{-1} and Latent heat of Fusion of water = 3.36×10^5 J kg^{-1})

(A) 35 g (B) 130 g (C) 20 g (D) 150 g

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Q70. JEE Main 2019 (Online) 12th January Morning Slot

A cylinder of radius R is surrounded by a cylindrical shell of inner radius R and outer radius 2R. The thermal conductivity of the material of the inner cylinder is K_1 and the of the outer cylinder is K_2. Assuming no loss of heat, the effective thermal conductivity of the system for heat flowing along the length of the cylinder is :

Open solution

(A) $K_1 + K_2$ (B) $\frac{K_1+3K_2}{4}$ (C) $\frac{K_1+K_2}{2}$ (D) $\frac{2K_1+3K_2}{5}$

Q71. JEE Main 2019 (Online) 11th January Evening Slot

Two rods A and B of identical dimensions are at temperature 30°C. If A is heated upto 180°C and B upto T°C, then the new lengths are the same. If the ratio of the coefficients of linear expansion of A and B is 4 : 3, then the value of T is

Open solution

(A) 200°C (B) 270°C (C) 230°C (D) 250°C

Q72. JEE Main 2019 (Online) 11th January Evening Slot

When 100 g of a liquid A at 100°C is added to 50 g of a liquid B at temperature 75°C, the temperature of the mixture becomes 90°C. The temperature of the mixture, if 100 g of liquid A at 100°C is added to 50 g of liquid B at 50°C, will be :

Open solution

(A) 60°C (B) 70°C (C) 85°C (D) 80°C

Q73. JEE Main 2019 (Online) 11th January Evening Slot

A thermometer graduated according to a linear scale reads a value x_0 when in contact with boiling water, and $x_0/3$ when in contact with ice. What is the temperature of an object in °C, if this thermometer in the contact with the object reads $x_0/2$?

Open solution

(A) 60 (B) 35 (C) 25 (D) 40

Q74. JEE Main 2019 (Online) 11th January Evening Slot

A metal ball of mass 0.1 kg is heated upto 500°C and dropped into a vessel of heat capacity 800 JK^{-1} and containing 0.5 kg water. The initial temperature of water and vessel is 30°C. What is the approximate percentage increment in the temperature of the water? [Specific Heat Capacities of water and metal are, respectively, 4200 $Jkg^{-1}\ K^{-1}$ and 400 $Jkg^{-1}\ K^{-1}$]

Open solution

(A) 20% (B) 25% (C) 15% (D) 30%

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Q75. JEE Main 2019 (Online) 11th January Morning Slot

Open solution

Ice at $-20^{o}C$ is added to 50 g of water at $40^{o}C$. When the temperature of the mixture reaches $0^{o}C$, it is found that 20 g of ice is still unmelted. The amount of ice added to the water was close to (Specific heat of water = 4.2J/g/^{o}C Specific heat of Ice = 2.1J/g/^{o}C Heat of fusion of water at $0^{o}C$= 334J/g)

(A) 100 g (B) 60 g (C) 50 g (D) 40 g

Q76. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

An unknown metal of mass 192 g heated to a temperature of $100^{o}C$ was immersed into a brass calorimeter of mass 128 g containing 240 g of water at a temperature of $8.4^{o}C$. Calculate the specific heat of the unknown metal if water temperature stabilizes at $21.5^{o}C$. (Specific heat of brass is 394 J kg^{-1} K^{-1})

(A) 458 J kg^{-1} K^{-1} (B) 1232 J kg^{-1} K^{-1} (C) 654 J kg^{-1} K^{-1} (D) 916 J kg^{-1} K^{-1}

Q77. JEE Main 2019 (Online) 10th January Morning Slot

Open solution

A heat source at $T = 10^3$ K is connected to another heat reservoir at $T = 10^2$ K by a copper slab which is 1 m thick. Given that the thermal conductivity of copper is 0.1 $WK^{-1}m^{-1}$, the energy flux through it in the steady state is -

(A) 200 Wm^{-2} (B) 65 Wm^{-2} (C) 120 Wm^{-2} (D) 90 Wm^{-2}

Q78. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

Temperature difference of $120^{o}C$ is maintained between ends of a uniform rod AB of length 2L. Another bent rod PQ, of same cross-section as AB and length $\frac{3L}{2}$, is connected across AB (see figure). In steady state, temperature difference between P and Q will be close to :



(A) $45^{o}C$ (B) $75^{o}C$ (C) $60^{o}C$ (D) $35^{o}C$

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Q79. JEE Main 2017 (Online) 8th April Morning Slot

In an experiment, a sphere of aluminium of mass 0.20 kg is heated upto 150°C. Immediately, it is put into water of volume 150 cc at 27°C kept in a calorimeter of water equivalent to 0.025 kg. Final temperature of the system is 40°C. The specific heat of aluminium is : (take 4.2 Joule = 1 calorie)

Open solution

(A) 378 J/kg $-^{o}$C (B) 315 J/kg $-^{o}$C (C) 476 J/kg $-^{o}$C (D) 434 J/kg $-^{o}$C

Q80. JEE Main 2017 (Offline)

A copper ball of mass 100 gm is at a temperature T. It is dropped in a copper calorimeter of mass 100 gm, filled with 170 gm of water at room temperature. Subsequently, the temperature of the system is found to be 75°C. T is given by: (Given : room temperature = 30°C, specific heat of copper = 0.1 cal/gm^{o}C)

Open solution

(A) 825°C (B) 800°C (C) 885°C (D) 1250°C

Q81. JEE Main 2016 (Online) 9th April Morning Slot

200 g water is heated from 40°C to 60°C. Ignoring the slight expansion of water, the change in its internal energy is close to (Given specific heat of water = 4184 J/kg/K) :

Open solution

(A) 8.4 kJ (B) 4.2 kJ (C) 16.7 kJ (D) 167.4 kJ

Q82. JEE Main 2016 (Online) 9th April Morning Slot

A simple pendulum made of a bob of mass *m* and a metallic wire of negligible mass has time period 2 s at T=0°C. If the temperature of the wire is increased and the corresponding change in its time period is plotted against its temperature, the resulting graph is a line of slope S. If the coefficient of linear expansion of metal is α then the value of S is :

Open solution

(A) α (B) $\frac{\alpha}{2}$ (C) 2α (D) $\frac{1}{\alpha}$

Q83. JEE Main 2016 (Offline)

A pendulum clock loses $12\ s$ a day if the temperature is $40^{\circ}C$ and gains $4\ s$ a day if the temperature is $20^{\circ}C$. The temperature at which the clock will show correct time, and the co-efficient of linear expansion (α) of the metal of the pendulum shaft are respectively :

Open solution

(A) $30^{\circ}C;\ \alpha = 1.85 \times 10^{-3}/^{\circ}C$ (B) $55^{\circ}C;\ \alpha = 1.85 \times 10^{-2}/^{\circ}C$

(C) $25^{\circ}C;\ \alpha = 1.85 \times 10^{-5}/^{\circ}C$ (D) $60^{\circ}C;\ \alpha = 1.85 \times 10^{-4}/^{\circ}C$

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Q84. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

An experiment takes 10 minutes to raise the temperature of water in a container from 0^oC to 100^oC and another 55 minutes to convert it totally into steam by a heater supplying heat at a uniform rate. Neglecting the specific heat of the container and taking specific heat of water to be 1 cal/g ^oC, the heat of vapourization according to this experiment will come out to be :

(A) 530 cal/g (B) 540 cal/g (C) 550 cal/g (D) 560 cal/g

Q85. JEE Main 2014 (Online) 19th April Morning Slot

Open solution

A black coloured solid sphere of radius R and mass M is inside a cavity with vacuum inside. The walls of the cavity are maintained at a temperature T_0. The initial temperature of the sphere is $3T_0$. If the specific heat of the material of the sphere varies as αT^3 per unit mass with the temperature T of the sphere, where α is a constant, then the time taken for the sphere to cool down to temperature $2T_0$ will be (σ is Stefan Boltzmann constant)

(A) $\frac{M\alpha}{4\pi R^2\sigma}\ln\left(\frac{3}{2}\right)$ (B) $\frac{M\alpha}{4\pi R^2\sigma}\ln\left(\frac{16}{3}\right)$
(C) $\frac{M\alpha}{16\pi R^2\sigma}\ln\left(\frac{16}{3}\right)$ (D) $\frac{M\alpha}{16\pi R^2\sigma}\ln\left(\frac{3}{2}\right)$

Q86. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

Water of volume 2 L in a closed container is heated with a coil of 1 kW. While water is heated, the container loses energy at a rate of 160J/s. In how much time will the temperature of water rise from 27^oC to 77^oC ? (Specific heat of water is 4.2 kJ/kg K and that of the container is negligible).

(A) 8 min 20 s (B) 6 min 2 s (C) 7 min (D) 14 min

Q87. JEE Main 2014 (Offline)

Open solution

Three rods of Copper, Brass and Steel are welded together to form a Y shaped structure. Area of cross - section of each rod $= 4cm^2$. End of copper rod is maintained at $100^\circ C$ where as ends of brass and steel are kept at $0^\circ C$. Lengths of the copper, brass and steel rods are 46, 13 and 12 cms respectively. The rods are thermally insulated from surroundings excepts at ends. Thermal conductivities of copper, brass and steel are $0.92, 0.26$ and 0.12 CGS units respectively. Rate of heat flow through copper rod is:

(A) 1.2 cal/s (B) 2.4 cal/s (C) 4.8 cal/s (D) 6.0 cal/s

Q88. JEE Main 2013 (Online) 23th April Morning Slot

Open solution

The ratio of the coefficient of volume expansion of a glass container to that of a viscous liquid kept inside the container is 1 : 4. What fraction of the inner volume of the container should the liquid occupy so that the volume of the remaining vacant space will be same at all temperatures?

(A) 2 : 5 (B) 1 : 4 (C) 1 : 64 (D) 1 : 8

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Q89. JEE Main 2013 (Online) 23th April Morning Slot

500 g of water and 100 g of ice at 0°C are in a calorimeter whose water equivalent is 40 g. 10 g of steam at 100°C is added to it. Then water in the calorimeter is : (Latent heat of ice = 80 cal/g, Latent heat of steam = 540 cal/g)

(A) 580 g (B) 590 g (C) 600 g (D) 610 g

Open solution

Q90. JEE Main 2013 (Online) 9th April Morning Slot

On a linear temperature scale Y, water freezes at -160°Y and boils at -50°Y. On this Y scale, a temperature of 340 K would be read as : (water freezes at 273 K and boils at 373 K)

(A) -73.7°Y (B) -233.7°Y (C) -86.3°Y (D) -106.3°Y

Open solution

Q91. AIEEE 2012 (Online) 19th May

A large cylindrical rod of length L is made by joining two identical rods of copper and steel of length $\left(\frac{L}{2}\right)$ each. The rods are completely insulated from the surroundings. If the free end of copper rod is maintained at $100^{\circ}\mathrm{C}$ and that of steel at $0^{\circ}\mathrm{C}$ then the temperature of junction is (Thermal conductivity of copper is 9 times that of steel)

(A) $90^{\circ}\mathrm{C}$ (B) $50^{\circ}\mathrm{C}$ (C) $10^{\circ}\mathrm{C}$ (D) $67^{\circ}\mathrm{C}$

Open solution

Q92. AIEEE 2012 (Online) 7th May

The heat radiated per unit area in 1 hour by a furnace whose temperature is $3000\ \mathrm{K}$ is $\left(\sigma = 5.7 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}\right)$

(A) $1.7 \times 10^{10}\ \mathrm{J}$ (B) $1.1 \times 10^{12}\ \mathrm{J}$ (C) $2.8 \times 10^{8}\ \mathrm{J}$ (D) $4.6 \times 10^{6}\ \mathrm{J}$

Open solution

Q93. AIEEE 2011

$100g$ of water is heated from $30^{\circ}C$ to $50^{\circ}C$. Ignoring the slight expansion of the water, the change in its internal energy is (specific heat of water is $4184\ J/kg/K$):

(A) $8.4\ kJ$ (B) $84\ kJ$ (C) $2.1\ kJ$ (D) $4.2\ kJ$

Open solution

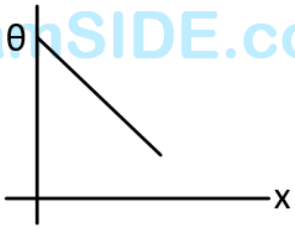
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Q94. AIEEE 2009

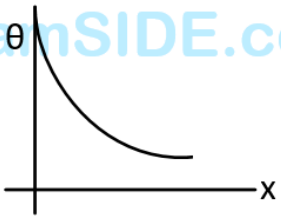
A long metallic bar is carrying heat from one of its ends to the other end under steady-state. The variation of temperature θ along the length x of the bar from its hot end is best described by which of the following figures?

Open solution

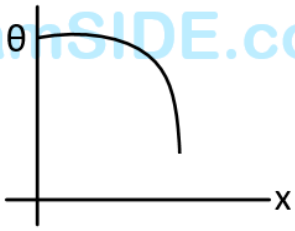
(A)



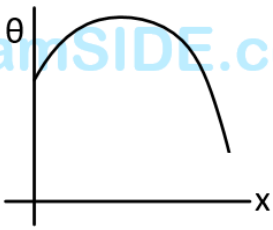
(B)



(C)



(D)



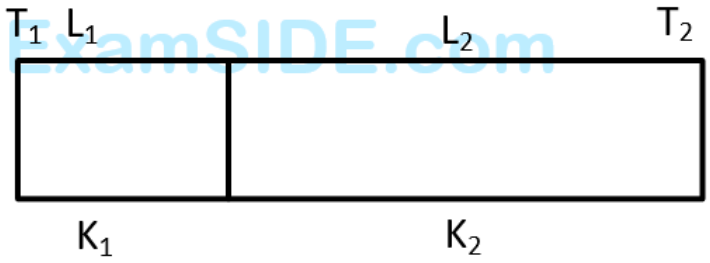
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Q95. AIEEE 2007

One end of a thermally insulated rod is kept at a temperature T_1 and the other at T_2. The rod is composed of two sections of length L_1 and L_2 and thermal conductivities K_1 and K_2 respectively. The temperature at the interface of the two section is

Open solution



(A) $\frac{(K_1L_1T_1+K_2L_2T_2)}{(K_1L_1+K_2L_2)}$

(B) $\frac{(K_2L_2T_1+K_1L_1T_2)}{(K_1L_1+K_2L_2)}$

(C) $\frac{(K_2L_1T_1+K_1L_2T_2)}{(K_2L_1+K_1L_2)}$

(D) $\frac{(K_1L_2T_1+K_2L_1T_2)}{(K_1L_2+K_2L_1)}$

Q96. AIEEE 2006

Assuming the Sun to be a spherical body of radius R at a temperature of TK, evaluate the total radiant power incident on Earth at a distance r from the Sun
Where r_0 is the radius of the Earth and σ is Stefan's constant.

Open solution

(A) $4\pi r_0^2R^2\sigma\frac{T^4}{r^2}$

(B) $\pi r_0^2R^2\sigma\frac{T^4}{r^2}$

(C) $r_0^2R^2\sigma\frac{T^4}{4\pi r^2}$

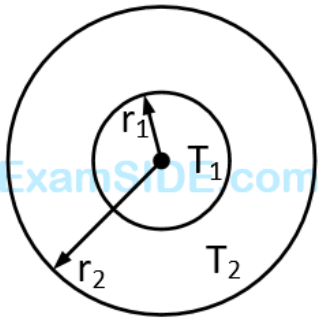
(D) $R^2\sigma\frac{T^4}{r^2}$

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Q97. AIEEE 2005

The figure shows a system of two concentric spheres of radii r_1 and r_2 are kept at temperatures T_1 and T_2, respectively. The radial rate of flow of heat in a substance between the two concentric spheres is proportional to

Open solution



(A) $In\left(\frac{r_2}{r_1}\right)$

(B) $\frac{(r_2-r_1)}{(r_1r_2)}$

(C) (r_2-r_1)

(D) $\frac{r_1r_2}{(r_2-r_1)}$

Q98. AIEEE 2004

If the temperature of the sun were to increase from T to $2T$ and its radius from R to $2R$, then the ratio of the radiant energy received on earth to what it was previously will be

Open solution

(A) 32

(B) 16

(C) 4

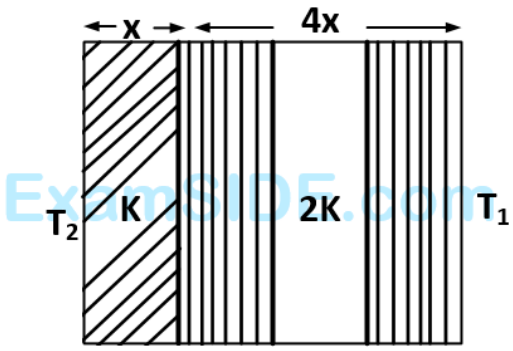
(D) 64

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Q99. AIEEE 2004

The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and $2K$ and thickness x and $4x$, respectively, are T_2 and T_1 $(T_2 > T_1)$. The rate of heat transfer through the slab, in a steady state is $\left(\frac{A(T_2-T_1)K}{x}\right)f$, with f equal to

Open solution



(A) $\frac{2}{3}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{1}{3}$

Q100. AIEEE 2004

Time taken by a $836\ W$ heater to heat one litre of water from $10^{\circ}C$ to $40^{\circ}C$ is

(A) $150\ s$ (B) $100\ s$ (C) $50\ s$ (D) $200\ s$

Open solution

Q101. AIEEE 2003

The earth radiates in the infra-red region of the spectrum. The spectrum is correctly given by

(A) Rayleigh Jeans law (B) Planck's law of radiation
(C) Stefan's law of radiation (D) Wien's law

Open solution

Q102. AIEEE 2002

Heat given to a body which raises its temperature by $1^{\circ}C$ is

(A) water equivalent (B) thermal capacity
(C) specific heat (D) temperature gradient

Open solution

Q103. AIEEE 2002

Which of the following is more close to a black body?

(A) black board paint (B) green leaves
(C) black holes (D) red roses

Open solution

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Q104. AIEEE 2002

Two spheres of the same material have radii $1\ m$ and $4\ m$ and temperatures $4000\ K$ and $2000\ K$ respectively. The ratio of the energy radiated per second by the first sphere to that by the second is

(A) $1:1$ (B) $16:1$ (C) $4:1$ (D) $1:9$

Open solution

Out of syllabus 16 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q105. JEE Main 2023 (Online) 29th January Morning Shift · **Numerical**

A body cools from 60°C to 40°C in 6 minutes. If, temperature of surroundings is 10°C. Then, after the next 6 minutes, its temperature will be _____°C.

Open solution

Q106. JEE Main 2022 (Online) 25th June Morning Shift · **Numerical**

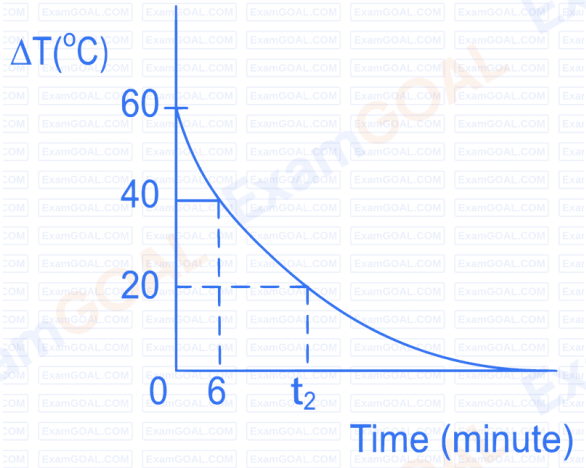
A steam engine intakes 50 g of steam at 100°C per minute and cools it down to 20°C. If latent heat of vaporization of steam is 540 cal g^{-1}, then the heat rejected by the steam engine per minute is _____ $\times 10^3$ cal.

(Given : specific heat capacity of water : 1 cal $g^{-1}\ {}^{\circ}C^{-1}$)

Open solution

Q107. JEE Main 2022 (Online) 24th June Evening Shift · **Numerical**

In an experiment to verify Newton's law of cooling, a graph is plotted between the temperature difference (ΔT) of the water and surroundings and time as shown in figure. The initial temperature of water is taken as 80°C. The value of t_2 as mentioned in the graph will be _____.



Open solution

Q108. JEE Main 2021 (Online) 22th July Evening Shift · **Numerical**

In 5 minutes, a body cools from 75°C to 65°C at room temperature of 25°C. The temperature of body at the end of next 5 minutes is _____° C.

Open solution

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Q109. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

A body cools from $80^{\circ}\mathrm{C}$ to $60^{\circ}\mathrm{C}$ in 5 minutes. The temperature of the surrounding is $20^{\circ}\mathrm{C}$. The time it takes to cool from $60^{\circ}\mathrm{C}$ to $40^{\circ}\mathrm{C}$ is :

(A) 500 s (B) $\frac{25}{3}$ s (C) 450 s (D) 420 s

Open solution

Q110. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

A body cools in 7 minutes from $60^{\circ}\mathrm{C}$ to $40^{\circ}\mathrm{C}$. The temperature of the surrounding is $10^{\circ}\mathrm{C}$. The temperature of the body after the next 7 minutes will be:

(A) $34^{\circ}\mathrm{C}$ (B) $28^{\circ}\mathrm{C}$ (C) $32^{\circ}\mathrm{C}$ (D) $30^{\circ}\mathrm{C}$

Open solution

Q111. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

A bowl filled with very hot soup cools from 98°C to 86°C in 2 minutes when the room temperature is 22°C. How long it will take to cool from 75°C to 69°C?

(A) 2 minutes (B) 0.5 minute (C) 1.4 minutes (D) 1 minute

Open solution

Q112. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

A body takes 4 min. to cool from 61° C to 59° C. If the temperature of the surroundings is 30° C, the time taken by the body to cool from 51° C to 49° C is :

(A) 4 min. (B) 3 min. (C) 8 min. (D) 6 min.

Open solution

Q113. JEE Main 2020 (Online) 3rd September Evening Slot · MCQ (Single Correct Answer)

A metallic sphere cools from 50°C to 40° in 300 s. If atmospheric temperature around is 20°C, then the sphere's temperature after the next 5 minutes will be close to :

(A) 35°C (B) 31°C (C) 33°C (D) 28°C

Open solution

Q114. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

A body takes 10 minutes to cool from 60°C to 50°C. The tempertature of surroundings is constant at 25°C. Then, the temperature of the body after next 10 minutes will be approximately :

(A) 47°C (B) 41°C (C) 45°C (D) 43°C

Open solution

Q115. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

Hot water cools from 60°C to 50°C in the first 10 minutes and 42°C in the next 10 minutes. The temperature of the surroundings is :

(A) 25°C (B) 10°C (C) 15°C (D) 20°C

Open solution

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Q116. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

A hot body, obeying Newton's law of cooling is cooling down from its peak value 80°C to an ambient temperature of 30°C. It takes 5 minutes in cooling down from 80°C to 40°C. How much time will it take to cool down from 62°C to 32°C ? (Given ℓn 2 = 0.693, ℓn 5 = 1.609)

Open solution

(A) 3.75 minutes (B) 8.6 minutes (C) 9.6 minutes (D) 6.5 minutes

Q117. JEE Main 2013 (Online) 25th April Morning Slot · MCQ (Single Correct Answer)

A mass of 50 g of water in a closed vessel, with surroundings at a constant temperature takes 2 minutes to cool from 30°C to 25°C. A mass of 100 g of another liquid in an identical vessel with identical surroundings takes the same time to cool from 30°C to 25°C. The specific heat of the liquid is : (The water equivalent of the vessel is 30 g.)

Open solution

(A) 2.0 kcal/kg (B) 7 kcal/kg (C) 3 kcal/kg (D) 0.5 kcal/kg

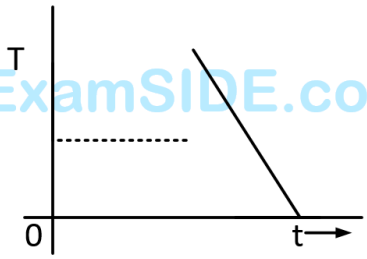
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Q118. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

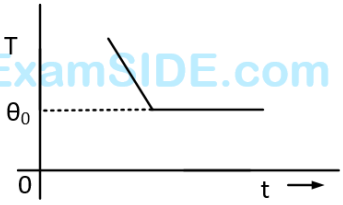
Open solution

If a piece of metal is heated to temperature θ and then allowed to cool in a room which is at temperature θ_0, the graph between the temperature T of the metal and time t will be closest to

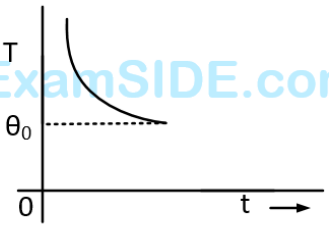
(A)



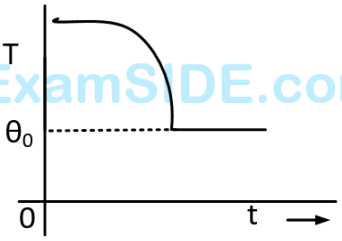
(B)



(C)



(D)



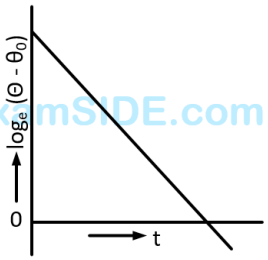
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Q119. AIEEE 2012 · MCQ (Single Correct Answer)

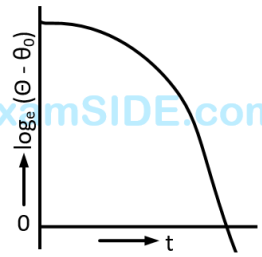
A liquid in a beaker has temperature $\theta(t)$ at time t and θ_0 is temperature of surroundings, then according to Newton's law of cooling the correct graph between $\log_e(\theta - \theta_0)$ and t is :

Open solution

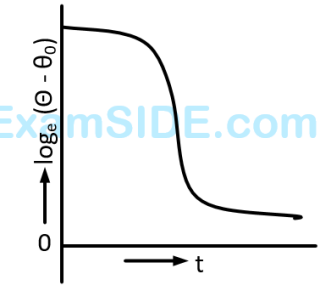
(A)



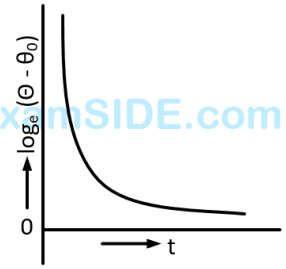
(B)



(C)



(D)



Q120. AIEEE 2003 · MCQ (Single Correct Answer)

According to Newton's law of cooling, the rate of cooling of a body is proportional to $(\Delta\theta)^n$, where $\Delta\theta$ is the difference of the temperature of the body and the surrounding, and n is equal to :

(A) two (B) three (C) four (D) one

Open solution

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Answer Key

Numerical

1. **922**	2. **360**	3. **5**	4. **40**	5. **40**
6. **300**	7. **313**	8. **12**	9. **90**	10. **60**
11. **21**	12. **42**	13. **2**	14. **20**	15. **50**
16. **40**	17. **60**			

MCQ (Single Correct Answer)

18. **D**	19. **C**	20. **B**	21. **C**	22. **C**
23. **D**	24. **C**	25. **A**	26. **D**	27. **A**
28. **A**	29. **B**	30. **D**	31. **D**	32. **C**
33. **B**	34. **B**	35. **C**	36. **A**	37. **A**
38. **D**	39. **D**	40. **C**	41. **A**	42. **B**
43. **D**	44. **D**	45. **B**	46. **B**	47. **C**
48. **C**	49. **B**	50. **A**	51. **A**	52. **C**
53. **B**	54. **A**	55. **B**	56. **A**	57. **A**
58. **A**	59. **D**	60. **A**	61. **B**	62. **B**
63. **B**	64. **B**	65. **A**	66. **C**	67. **A**
68. **B**	69. **C**	70. **B**	71. **C**	72. **D**
73. **C**	74. **A**	75. **D**	76. **D**	77. **D**
78. **A**	79. **D**	80. **C**	81. **C**	82. **A**
83. **C**	84. **C**	85. **C**	86. **A**	87. **C**
88. **B**	89. **B**	90. **C**	91. **A**	92. **A**
93. **A**	94. **A**	95. **D**	96. **B**	97. **D**
98. **D**	99. **D**	100. **A**	101. **D**	102. **B**
103. **A**	104. **A**			

Out of syllabus

105. **28**	106. **31**	107. **16**	108. **57**	109. **A**
110. **B**	111. **C**	112. **D**	113. **C**	114. **D**
115. **B**	116. **B**	117. **D**	118. **C**	119. **A**
120. **D**				

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